# SIH 2026 — Problem Statement Decision Guide

Detailed comparison of the 9 PSs shared • Hackathon execution, AI potential, demo impact, risk, and recommended tech stacks

## Executive Ranking

| Rank | PS | Problem | AI / Tech Depth | MVP Feasibility | Demo Impact | Differentiation | Key Risk | Overall |
|---|---|---|---|---|---|---|---|---|
| 1 | 26106 | Email Threat Intelligence | ★★★★★ | ★★★★ | ★★★★★ | ★★★★★ | Forensic attribution limits | 9.3/10 |
| 2 | 26107 | BIS AI Assistant | ★★★★★ | ★★★★★ | ★★★★★ | ★★★★★ | Hallucination / source grounding | 9.3/10 |
| 3 | 26034 | Package Compliance | ★★★★★ | ★★★★ | ★★★★★ | ★★★★★ | Legal-rule accuracy | 9.2/10 |
| 4 | 26103 | Infrastructure Prediction | ★★★★★ | ★★ | ★★★★★ | ★★★★★ | Historical data access | 9.0/10 |
| 5 | 26115 | Medical Waste Robot | ★★★★ | ★★ | ★★★★★ | ★★★★★ | Hardware + CAD execution | 8.9/10* |
| 6 | 26033 | Farmer Marketplace | ★★★★ | ★★★★ | ★★★★ | ★★★★ | Marketplace differentiation | 8.5/10 |
| 7 | 26044 | Academia–Industry | ★★★★ | ★★★ | ★★★★ | ★★★ | Huge scope | 8.2/10 |
| 8 | 26075 | Capacity Connect | ★★★ | ★★★★★ | ★★★ | ★★ | Common LMS pattern | 7.5/10 |
| 9 | 26032 | Farmer Queue | ★★ | ★★★★★ | ★★★ | ★★ | Low technical novelty | 7.2/10 |

*26115 assumes a strong robotics/CAD/electronics team; for a software-only team, its suitability drops substantially.

### Recommended shortlist for a software + AI team

■ **26106** — Best cybersecurity + AI story. ■ **26107** — Best LLM/RAG balance of innovation and buildability. ■ **26034** — Best computer-vision + rules demo. ■ **26103** — Highest ML ceiling, but highest data risk.

# Core Product & Hackathon Comparison

| PS | Primary users | Core workflow | Main AI | Best demo moment | Build risk |
|---|---|---|---|---|---|
| 26106 | Security analysts / investigators | Upload .eml → parse → threat score → header/IP intelligence → graph → report | NLP + ML + anomaly/rule analysis | Upload one email and instantly show risk, indicators, trace graph and report | Medium |
| 26107 | MSMEs, industries, consumers | Question → retrieve BIS source → reason → answer + clause citations → roadmap | LLM + RAG + embeddings | Ask a product question and receive an official, cited answer | Low–Medium |
| 26034 | Legal Metrology inspectors | Product photos → OCR/CV → extract fields → rules → highlighted violations → report | Vision + OCR + rule engine | Highlight missing/incorrect label declarations directly on the package | Medium |
| 26103 | MoSPI / project officials | Historical project data → features → predictions → risk score → alerts | ML + forecasting + explainability | Select a project and show predicted cost/delay risk with drivers | High |
| 26115 | Hospitals / waste operators | Camera → classify → navigate/collect → segregate → track | Computer vision + robotics | Physical robot identifies and sorts waste | Very High |
| 26033 | Farmers/FPOs + buyers | Listings → demand forecast → buyer match → order aggregation → route | Forecasting + optimization | Show a farmer order being matched and routed at a better realization | Medium |
| 26044 | Students + industry + academia | Assessment → skill profile → gap → learning → internship/placement match | Recommendation + matching | Student gets a personalized path from skill gap to internship | Medium–High |
| 26075 | Trainees + trainers + admins | Profile → course → content → assessment → certificate → analytics | Competency mapping + recommendation | Competency engine recommends trainer/course for a skill gap | Low–Medium |
| 26032 | Farmers + procurement centres | Register → slot → token → live queue → procurement → payment | Wait prediction / optimization | Live queue shows current token and predicted waiting time | Low |

# Hackathon-Ready Tech Stacks — Software PSs

The stacks below intentionally avoid over-engineering. The goal is a fast, reliable MVP with enough depth for judging.

| PS | Frontend | Backend | AI / Core | Data / Storage | APIs / Tools | Deployment |
|---|---|---|---|---|---|---|
| 26106 | Next.js + TypeScript + Tailwind + shadcn/ui | FastAPI + Python | scikit-learn/XGBoost + Hugging Face; hybrid rules | PostgreSQL / Supabase; object storage | VirusTotal, AbuseIPDB, IP geolocation, dnspython, Python email parser, React Flow, Leaflet | Vercel + Render/Railway |
| 26107 | Next.js + TypeScript + Tailwind + shadcn/ui | FastAPI or Next.js API routes | LLM + embeddings + RAG; LangChain/LlamaIndex optional | Supabase Postgres + pgvector | PyMuPDF, document chunking, BIS-authorized knowledge sources | Vercel + Render/Railway |
| 26034 | Next.js + TypeScript + Tailwind + shadcn/ui | FastAPI + Python | PaddleOCR/Tesseract + OpenCV + YOLO optional; deterministic rules | PostgreSQL / Supabase + image storage | ReportLab, image annotation, rule JSON/YAML | Vercel + Render/Railway |
| 26103 | Next.js + TypeScript + Tailwind + Recharts | FastAPI + Python | XGBoost/LightGBM + statistical baselines + SHAP | PostgreSQL / Parquet; Pandas | scikit-learn, Plotly/Recharts, model evaluation | Vercel + Render/Railway |
| 26033 | Next.js + TypeScript + Tailwind + shadcn/ui | FastAPI / Node.js | XGBoost/Prophet-style forecasting + optimization | PostgreSQL / Supabase | OSRM/OpenStreetMap or routing API; maps with Leaflet | Vercel + Render |
| 26044 | Next.js + TypeScript + Tailwind + shadcn/ui | Next.js API / FastAPI | Skill embeddings + recommendation/matching | PostgreSQL / Supabase + storage | Resume parser, embeddings, optional LLM | Vercel + Supabase |
| 26075 | Next.js + TypeScript + Tailwind | Next.js API / FastAPI | Competency scoring + recommendation | Supabase Postgres + storage | Supabase Auth; PDF certificate generation | Vercel + Supabase |
| 26032 | Next.js / React Native + TypeScript | Node.js or FastAPI | Queue prediction + slot optimization | PostgreSQL / Supabase | SMS provider, QR/token generation | Vercel + Render/Supabase |

# SIH26115 — Robotics Stack

| Layer | Hackathon choice | Purpose |
|---|---|---|
| CAD / Product | Autodesk Fusion | Mandatory design environment; assembly, simulation, motion study, renders |
| Vision | Python + OpenCV + YOLO | Identify waste categories from camera frames |
| Robot control | ESP32 / Raspberry Pi | Sensor, motor and actuator control; choose based on prototype |
| Navigation | ROS 2 optional; simpler state machine for MVP | Autonomous movement / obstacle handling |
| Sensors | Ultrasonic / ToF + encoders | Obstacle detection and movement feedback |
| Actuation | Servo / geared motors / conveyor or flap mechanism | Collection and segregation |
| Backend | FastAPI | Telemetry and waste-event APIs |
| Dashboard | Next.js + TypeScript | Waste logs, battery, compartment fill, alerts |
| Database | Supabase / PostgreSQL | Collection history and device data |
| Deployment | Local edge device + cloud dashboard | Keep critical control local; sync analytics to cloud |

**Important:** 26115 is not a normal software-only hackathon. Autodesk's PS specifically expects Fusion-based product development, simulation/motion/exploded views and a manufacturable prototype direction. Choose it only if the team can execute the physical design.

# What to Build for the Demo — MVP → Differentiator

| PS | MVP (must work) | Differentiator (add after MVP) |
|---|---|---|
| 26106 | Upload .eml; header parsing; SPF/DKIM/DMARC; URL/domain checks; risk score; report | Threat graph + campaign clustering + confidence-based origin trace |
| 26107 | Chat; document retrieval; answer with citations; product→standard lookup | Certification roadmap + multilingual + clause comparison + guided agent workflow |
| 26034 | Upload package images; OCR; required-field detection; rule checks; report | Multi-side package fusion + visual violation overlays + e-commerce listing scan |
| 26103 | Historical dataset; baseline model; cost/delay predictions; risk dashboard | Explainable risk drivers + scenario simulation + intervention recommendations |
| 26115 | Prototype sorting mechanism; vision classification; basic movement; tracking | Autonomous navigation + optimized segregation mechanism + Fusion simulation |
| 26033 | Farmer/buyer listings; orders; matching; simple logistics | Demand forecasting + route optimization + price realization calculator |
| 26044 | Skill assessment; profile; skill gap; recommendation; internship matching | Adaptive learning path + verified portfolio + institution analytics |
| 26075 | Auth; course; content; MCQ; trainer/admin dashboards | Competency engine + personalized learning/trainer recommendations |
| 26032 | Registration; slot booking; token; live queue; status | Wait-time prediction + dynamic slot allocation + centre load balancing |

# Final Selection Strategy

**If your team is software + AI:** shortlist 26106, 26107, 26034.
**If your team is ML/data-heavy and can access strong historical data:** consider 26103.
**If you have genuine robotics/CAD talent:** 26115 becomes a top-tier option.
**If you want a safer, more conventional build:** 26033 or 26075.
**If finishing reliably is the priority:** 26032.

**Hackathon rule of thumb:** Build one complete story end-to-end. A working pipeline with measurable outcomes, explainable AI, polished UX and a credible deployment plan is stronger than a large list of half-finished features.